

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

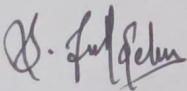
Annexure A

Dry Run Evaluation

Code of Conduct:

I Tamil Selvan Sasikumaran (192524134) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:







Student 38: Tamil Selvan Sasikumar | Register Number: 192524134

1. Question 12 - Local Search / Optimization

A delivery company has routes $R1=120$ km, $R2=95$ km, $R3=105$ km and $R4=85$ km. The current solution is $R1$

Perform a local-search improvement and explain local optimum versus global optimum.

2. Question 13 - Online Search Agent

A robot is placed in an unfamiliar building. Initially it knows only $A \rightarrow B$ and $A \rightarrow C$. Later it discovers $B \rightarrow D$ and $D \rightarrow G$

Trace one possible online-search route and show how the agent updates its knowledge.

3. Question 14 - Constraint Satisfaction Problem

Four students A, B, C and D must be assigned to 9 AM, 10 AM and 11 AM.

Constraints: A not at 9 AM, B before C, D not at the same time as A; C not at 11 AM.

Identify variables, domains and constraints, then find one valid assignment.

Student	Domain	Constraints	Assigned Slot
A	9/10/11 AM		
B	9/10/11 AM		
C	9/10/11 AM		
D	9/10/11 AM		

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy	Understands the problem with minor misunderstandings	Demonstrates partial understanding of the search problem	Shows little or no understanding of the search problem
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors	Performs most iterations correctly with minor mistakes in state expansion or data structure updates	Performs only some iterations correctly, several errors in tracing or node expansion	Unable to correctly perform the dry run or expand states
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules	Applies the algorithm correctly with a few procedural errors	Applies only basic algorithm steps with noticeable mistakes	Fails to apply the correct search algorithm
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification	Solution path is mostly correct with minor errors	Partially correct solution path, goal may not be reached correctly	Incorrect or incomplete solution path
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented	Presentation is clear with minor organizational issues	Limited explanation and organization, reasoning is partially clear	Poor presentation with unclear or missing reasoning

20 + 25 + 15 + 15 + 9

85

7

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Course name: Artificial Intelligence
Course code: CSEA1710

C02-AT3

1) local search / optimization

Given:

$$R_1 = 120 \text{ km}$$

$$R_2 = 95 \text{ km}$$

$$R_3 = 105 \text{ km}$$

$$R_4 = 85 \text{ km}$$

$$\text{current solution} = R_1 = 120 \text{ km}$$

local search improvement:

Compare the neighboring routes:

$$R_2 = 95 \text{ km} \rightarrow \text{improvement of } 25 \text{ km}$$

$$R_3 = 105 \text{ km} \rightarrow \text{improvement of } 15 \text{ km}$$

$$R_4 = 85 \text{ km} \rightarrow \text{improvement of } 35 \text{ km}$$

So, choose $R_4 = 85 \text{ km}$

Final solution: $R_4 = 85 \text{ km}$

local optimum vs Global optimum

- > local optimum: A solution that is better than its immediate neighboring solutions.
- > global optimum: The best solution among all possible solutions.

2) online search agent

Initial knowledge:

- $A \rightarrow B$
- $A \rightarrow C$

The robot starts from A.

One possible online-search route:

$A \rightarrow B \rightarrow D \rightarrow G$

knowledge updates:

step	location	new knowledge
1	A	$A \rightarrow B, A \rightarrow C$
2	B	Discovers $B \rightarrow D$
3	D	Discovers $D \rightarrow G$
4	G	Goal reached.

The agent continuously explores the environment, discovers new connections, and updates its knowledge while moving.

constraint satisfaction Problem:

variables:

A, B, C, D represent the four students.

Domains

Each student can be assigned:

$\{9 \text{ AM}, 10 \text{ AM}, 11 \text{ AM}\}$

constraints

$\rightarrow A \neq 9 \text{ AM}$

$\rightarrow B$ must be before C

$\rightarrow D \neq A$

$\rightarrow C \neq 11 \text{ AM}$

One valid assignment:

Student	Domain	Constraint	Assigned Slot
A	9/10/11 AM	$A \neq 9 \text{ AM}$	10 AM
B	9/10/11 AM	B before C	9 AM
C	9/10/11 AM	$C \neq 11 \text{ AM}$	10 AM
D	9/10/11 AM	$D \neq A$	11 AM

verification:

$\rightarrow A = 10 \text{ AM} \rightarrow \text{not } 9 \text{ AM}$

$\rightarrow B = 9 \text{ AM}, C = 10 \text{ AM} \rightarrow B$ is before C

$\rightarrow D = 11 \text{ AM}, A = 10 \text{ AM} \rightarrow$ different times

$\rightarrow C = 10 \text{ AM} \rightarrow \text{not } 11 \text{ AM}$

Therefore $A = 10 \text{ AM}, B = 9 \text{ AM}, C = 10 \text{ AM}, D = 11 \text{ AM}$ is a valid assignment.